

MATH 301: Homework 2

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Question 1: Section 2.1-10

Section 2.3: q4

Section 2.3: q5

Section 2.3: q6

Section 2.3: q8

Section 2.3: q10

Section 2.4: q2

If $S := \{\frac{1}{n} - \frac{1}{m} | n, m \in \mathbb{N}\}$ find $\inf S$ and $\sup S$.

Consider for the following:

$$n \in \mathbb{N} \implies 0 < \frac{1}{n} \leq 1$$

$$m \in \mathbb{N} \implies -1 < -\frac{1}{m} \leq 0$$

Finding Supremum

Find the maximum value of $\frac{1}{n}$ and minimum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \implies \frac{1}{n} = 1 \\ \implies -\frac{1}{m} = 0 \end{cases}$$

Note $\sup S = 1$ this statement holds if 1 the smallest upperbound of S .

Consider: $\lim_{m \rightarrow \infty} (1 - \frac{1}{m})$

$$|(1 - \frac{1}{m}) - 1| < \epsilon \iff |-\frac{1}{m} + (1 - 1)| < \epsilon \iff |-\frac{1}{m}| < \epsilon \iff |\frac{1}{m}| < \epsilon \iff \frac{|1|}{m} < \epsilon$$

$$\leftarrow m > \frac{1}{\epsilon} \leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{m \rightarrow \infty} (1 - \frac{1}{m}) = 1$$

Consider: $\lim_{n \rightarrow \infty} (\frac{1}{n} - 1)$

$$|(\frac{1}{n} - 1) + 1| < \epsilon \iff |\frac{1}{n} + (1 - 1)| < \epsilon \iff |\frac{1}{n}| < \epsilon \iff \frac{|1|}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

$$\leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{n \rightarrow \infty} (\frac{1}{n} - 1) = -1$$

Thus, we can say that the $\sup S = 1$.

Finding Infimum

Find the minimum value of $\frac{1}{n}$ and maximum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \implies \frac{1}{n} = 0 \\ \implies -\frac{1}{m} = -1 \end{cases}$$

Consider: $\lim_{n \rightarrow \infty} (\frac{1}{n} - 1)$

Section 2.4: q7

Let A, B be bounded nonempty subsets of $\mathbb{R}$, and let $A + B := \{a + b \mid a \in A, b \in B\}$.

Prove $\sup(A + B) = \sup A + \sup B$ and $\inf(A + B) = \inf A + \inf B$.

Consider for the following:

$$\sup A = a_s \implies \forall a \in A, a < a_s$$

$$\sup B = b_s \implies \forall b \in B, b < b_s$$

Observe:

$$\begin{array}{c} (a < a_s) \\ + (b < b_s) \\ \hline a + b < a_s + b_s \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$

$$\text{Therefore, } \sup A + B = a_s + b_s$$

$$\inf A = a_i \implies \forall a \in A, a > a_i$$

$$\inf B = b_i \implies \forall b \in B, b > b_i$$

Observe:

$$\begin{array}{c} (a > a_i) \\ + (b > b_i) \\ \hline a + b > a_i + b_i \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$

$$\text{Therefore, } \inf A + B = a_i + b_i$$

Section 2.4: q9

Let $X = Y := \{x \in \mathbb{R} : 0 < x < 1\}$. Define $h : X \times Y \rightarrow \mathbb{R}$ by $h(x, y) := 2x + y$.

(a) For each $x \in X$, find $f(x) := \sup\{h(x, y) : y \in Y\}$; then find $\inf\{f(x) : x \in X\}$.

(b) For each $y \in Y$, find $g(y) := \inf\{h(x, y) : x \in X\}$; then find $\sup\{g(y) : y \in Y\}$.

Compare with the result found in part (a).

Item (a)

$$\forall x \in X, f(x) := \sup\{h(x, y) : y \in (0, 1)\} \implies f(x) = 2x + 1$$

$$f(x) = 2x + 1 \wedge x \in (0, 1) \implies \inf\{h(x, y) : x \in X\} = 2(0) + 1 = 1$$

Item (b)

$$\begin{aligned}\forall y \in Y, g(i) := \inf\{h(x, y) : x \in (0, 1)\} &\implies g(y) = 2(0) + y = y \\ g(y) = y \wedge y \in (0, 1) &\implies \sup\{g(y) : y \in Y\} = 1\end{aligned}$$

Section 2.4: q14

If $y > 0$ show $\exists n \in \mathbb{N} \ni \frac{1}{2^n} < y$

Recall:

$$a > b \implies \frac{1}{a} < \frac{1}{b}$$

Observe:

$$\begin{aligned}P(n) : n &< 2^n \\ P(1) : 1 &< 2^1 \\ P(k) : k &< 2^k \\ 2k &< 2^{k+1} \\ 2k + 1 &< 2k < 2^{k+1} \\ \therefore P(n) &\text{ holds } \forall n \in \mathbb{N}\end{aligned}$$

Thus we can say $\frac{1}{2^n} < \frac{1}{n}$. By Archimedian Property as $y > 0 \implies \frac{1}{n} \leq y$

$$\frac{1}{2^n} < \frac{1}{n} \leq y$$

Section 2.4: q15

Modify the argument in Theorem 2.4.7 to show that $\exists y > 0 \in \mathbb{R}$ s.t. $y^2 = 3$

Theorem 2.4.7 $\exists x > 0 \in \mathbb{R} \ni x^2 = 2$

$$S = \{s \in \mathbb{R} \mid s \geq 0 \text{ and } s^2 < 3\}$$

Existence of Supremum

$$0^2 = 0 < 3 \implies 0 \in S \implies S \neq \emptyset$$

Let $\forall s \in S, s < 3$ Suppose $\exists s \in S \ni s \geq 3 \implies s^2 \geq 9 > 3 \nmid \implies S$ is bounded above.

$S \neq \emptyset \wedge S$ is bounded above $\implies \exists x \in \mathbb{R} \ni x = \sup S$ (By Completeness Property.)

Trichotomy

By Trichotomy, one and only one, of the following must be true:

$$(x^2 < 3) \quad (x^2 > 3) \quad (x^2 = 3)$$

(Case: $(x^2 > 3)$)

Suppose for the sake of contradiction x is the least upper bound of S

$$\text{Show: } \forall n \in \mathbb{N}, (x - \frac{1}{n})^2 > 3 \implies \forall s \in S, (x - \frac{1}{n})^2 > s^2 \implies (x - \frac{1}{n}) > s$$

Observe the following:

$$(x - \frac{1}{n})^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2}$$
$$x^2 - \frac{2x}{n} + \frac{1}{n^2} \xrightarrow{\text{Dropping } \frac{1}{n^2}} (x - \frac{1}{n})^2 > x^2 - \frac{2x}{n}$$

To hold supposition, the inequality must be strictly greater than three.

$$x^2 - \frac{2x}{n} > 3$$
$$x^2 - 3 > \frac{2x}{n}$$
$$\frac{1}{x^2 - 3} < \frac{n}{2x}$$
$$\frac{2x}{x^2 - 3} < n$$

Since $x^2 > 3 \implies (x^2 - 3) > 0 \implies \exists n \ni x > \frac{1}{n} > 0$.

$$(x - \frac{1}{n})^2 > 3 \implies \forall s \in S, s^2 < 3 < (x - \frac{1}{n})^2$$
$$\implies x - \frac{1}{n} \text{ is an upper bound of } S$$

$$x - \frac{1}{n} < x \implies x \text{ is not the least upper bound of } S \nmid$$

$$\therefore (x^2 \not> 3)$$

(Case: $(x^2 < 3)$)

Observe the following:

$$(x + \frac{1}{n})^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2}$$

Note $\forall n \in \mathbb{N}, \frac{1}{n^2} \leq \frac{1}{n}$

$$x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}$$
$$\implies x^2 + \frac{2x+1}{n} < 3$$
$$\implies \frac{2x+1}{n} < 3 - x^2$$
$$\implies \frac{n}{2x+1} > \frac{1}{3-x^2}$$
$$\implies n > \frac{2x+1}{3-x^2}$$

As $x^2 < 3 \implies 3 - x^2 > 0$. Therefore by Archimedian Property $\implies (x + \frac{1}{n}) \in S$
 $\implies x$ is not an upperbound $\nless$.

$$\therefore (x^2 \nless 3)$$

As $x^2 \nless 3$ and $x^2 \nless 3$. By Trichotomy $x^2 = 3$